

Documentation

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Preface

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1 Clinical Core Methods

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1.1 Background

This document provides information about the ABC-DS study and documentation of data currently available on the LONI website. The initial ABC-DS study began in 2015. Funded by the National Institutes on Health (NIH) through the **U01** mechanism, from 2015-2020, ABC-DS was the combined/harmonized effort of two groups of investigators: 1) the Alzheimer's Dementia in Down Syndrome (**ADDS**) group (Columbia University/NYS Institute for Basic Research in Developmental Disabilities, Massachusetts General Hospital, University of California Irvine); 2) the Neurodegeneration in Aging Down Syndrome (**NiAD**) group (University of Pittsburgh, University of Wisconsin Madison, University of Cambridge, Banner Institute, Washington University).

From 2020-2026, the ABC-DS study was funded by NIH through the **U19** mechanism. The number of participating institutions increased to ten (Columbia University/NYS Institute for Basic Research in Developmental Disabilities, Massachusetts General Hospital, University of California Irvine, University of Pittsburgh, University of Wisconsin Madison, University of Cambridge, Washington University, Kansas University Medical Center, and the University of Puerto Rico). A single, harmonized protocol, including many of the ADDS and NiAD measures, was adopted by all ten institutions. The U19 project includes new participants as well as individuals continuing from the U01 projects.

While the focus of ABC-DS is on individuals with Down Syndrome, NiAD also recruited sibling controls, and unaffected siblings continue to be recruited in the U19. In the ongoing U19 study, in addition to new participants, as many former U01 participants as possible have been recruited, for continued evaluation.

ABC-DS participants are scheduled to be seen every 16 months. Sibling controls receive a shorter cognitive evaluation battery than participants with Down Syndrome and do not receive caregiver assessments.

1.2 Available Data

ABC-DS data has been harmonized and made available to the research community through the Laboratory of Neuroimaging website (LONI). Cumulative data released at annual intervals include evaluations in which a consensus diagnosis has been generated as of a set date, based on a defined battery of tests. Currently, data on a total of 1,274 evaluations conducted on 503 participants with Down Syndrome (cases) are available on LONI; on 503 individuals with Down Syndrome and 59 sibling controls, along with 161 evaluations conducted on 59 sibling controls. Case and control data are provided in separate tables.

Data collected at multiple evaluations are organized in a one-record-per-evaluation format sorted by participant ID, subject label, and consecutive evaluation (event_sequence), regardless of study of recruitment. For example, a participant with three evaluations in the U01 and one evaluation in the U19 would have data for event_sequence 1, 2, 3 and 4 (the first U19 evaluation). A participant seen in the first and third U01 evaluation with one evaluation in the U19 would have data for event_sequence 1,2 and 3. Note that event_sequence is numbered consecutively, regardless of whether a particular 16 month month evaluation (in this case the second U01 evaluation) is missed. The age_at_event_and_latency table provides the age of the participant at each evaluation and the latency in days from baseline of each subsequent evaluation. The number of evaluations for which data are currently available is shown in Table 1. Please note that the N of evaluations available for any specific measure is affected by a number of factors (please see descriptions of individual measures, below).

Table 1. Number of available ABC-DS evaluations

Evaluation (event_sequence)	Down Syndrome Cases	Sibling Controls
1st (baseline)	503	59
2nd evaluation	346	35
3rd evaluation	283	17
4th evaluation	138	5
5th evaluation	4	1
Total evaluations	1,274	116

1.3 ABC-DS Cognitive & Caregiver Measures (2015-2026)

Some measures in the harmonized data available on LONI were collected in the U19 study, but not in one or both of the U01 studies (ADDS and/or NiAD); for the tables containing these measures (flagged in the descriptions below), sample sizes will be smaller than for measures administered in all three studies.

1.4 Measures in the ABC-DS Integrated Dataset

1.4.1 Cognitive Assessment Measures

For all the cognitive measures, before performing any modelling, please check the distributions of the variables. The vast majority of them have a skewed distributions and are zero inflated, which requires either some type of transformation or proper modelling.

- a) The **Down Syndrome Mental Status Examination (DSMSE)** (an enhanced version)(Haxby 1989) provides an omnibus measure of neuropsychological functioning, assessing recall of personal information, orientation, immediate and delayed memory, language, visuospatial function and praxis.
- b) The Rapid Assessment for Developmental Disabilities (RADD)(McCarthy 1972) Assesses the ability of the individual to copy simple hand movements. For ABC-DS we are only using the Hand Movements and Imitation Subtest of the measure.²
- c) The McCarthy Scales of Children's Abilities: Category Fluency Test(Korkman et al. 2007) An adaptation of the McCarthy Category Fluency Test has been employed to assess changes in language ability and frontal lobe function over time. It requires participants to name as many foods and animals as they can within 20 seconds. We have been using a single summary score generated by adding the number of correct responses for each category.²
- d) The NEPSY (2nd ed.) Word Generation Semantic Fluency Subtest(Korkman et al. 2007) assesses verbal productivity through the ability to generate as many words as possible in 20 sec. within a specific semantic category (e.g., animals, food and drinks). The score is the total number of unique animal names/foods/drinks generated.²
- e) Kaufman Brief Intelligence Test (2nd ed)(Kaufman 2004) The KBIT-2 is a brief measure of verbal and nonverbal intelligence for ages 4-90. It provides crystallized (verbal) and Fluid (nonverbal) score along with an IQ composite score.¹
- f) The Wechsler Intelligence Scale for Children, 4th Edition (WISC-IV)(Wechsler 2003) Block Design subtest was used to measure visuo-spatial organization. The subtest is a core perceptual reasoning timed test that requires that the individual reproduce designs using blocks that have red surfaces, white surfaces, and surfaces divided diagonally into half red and half white. The Digit span forward subtest of the WISC-IV was used to assess rote learning and memory. The WISC-IV version is somewhat easier (as it is designed for children ages 6 to 16). It contains series ranging in length from two to nine digits and primarily involves rote learning and memory. Digit forward requires participants to recite a series of digits in the exact order given by the evaluator (forward). Conversely, Digit Span backwards requires participants to recite series of digits backwards and taps ability to manipulate verbal material in working memory.

- g) Haxby Extension.(Haxby 1989) Like the WISC Block Design subtest, the Haxby Extension requires that participants reproduce designs using blocks with red surfaces, white surface, and surfaced divided diagonally into half red and half white. However, the designs are simpler than those on the WISC, such as starting with reproducing a single block. Possible cores range from 0-16.
- h) The Beery-Buktenica Developmental Test of Visual Motor Integration (VMI)(Beery 2004) involves the copying of figures of progressive complexity using paper and pencil. It is designed to assess the extent to which the individual can integrate his/her visual and motor abilities. Scores range from 0 to 30 (higher scores indicating more correctly copied items).
- i) The Modified Cued Recall Test (mCRT)(Devenny et al. 2002) is a list-learning task that measure verbal learning and memory. The mCRT consists of a training period in which 12 items are presented, 4 at a time, with each item accompanied by a unique category cue. The testing phase consists of 3 trials which generates a free recall score (FRS -spontaneous recall of the 12 items), cued recall score (CRS-items only recalled when the category cue is provided), a total score (sum FRS and CRS), and an intrusion score (items named that were not on the original list).
- j) The Modified Cats and Dogs Task(Ball et al. 2008) is an adapted version of the Stroop Task and it assesses cognitive inhibition. Specifically, the task measures an individual's ability to inhibit a natural response (naming pictures of a dog as "dog") and to replace it with an incongruent one (naming pictures of a dog as "cat").
- k) The Purdue Pegboard Test(Tiffin and Asher 1948) measures manual dexterity (left and right hand separately) as a participant places pins/pegs in a pegboard with 2 parallel rows.
- l) The Cancellation Task-Revised(Krinsky-McHale et al. 2008) measures processing speed, visual selective attention, vigilance, perceptual speed and visuomotor ability by asking participants to circle a target stimulus/picture on a page containing both target pictures and distractor pictures.
- m) The Tinetti Assessment Tool: Gait subtest(Tinetti 1986) quantifies gait in 7 domains by observing participants while walking. Those domains are: Initiation of Gait, Step Length, Step Height/ Foot Clearance, Step Symmetry, Step Continuity, Path, Trunk and Walking Stance.

Caregiver Interview Measures

- a) The Dementia Questionnaire for People with Learning Disabilities (DLD).(Eurlings et al. 2006) The DLD consists of 50 items in 8 subscales, including short-term memory, long-term memory spatial and temporal orientation, speech, practical skills, mood, activity and interest, and behavioral disturbances. The DLD generates two scores, one reflecting cognitive abilities (DLD/SCS) and one reflecting social skills (DLD/SOS).

- b) The NTG Early Detection Screen for Dementia (NTG-EDSD).(Esralew et al. 2017)
The NTG-EDSD includes sections on behavioral/cognitive domains that could reflect indications of dementia clinical progression. In the current ABC-DS cohort, the DLD and NTG significantly differentiated participants who were Unaffected from those with MCI-DS and Dementia.
- c) The Vineland Adaptive Behavior Scales, 3rd edition (VABS-3),(Sparrow et al. 2016)
Domain Level Parent/ Caregiver Form, provides a standardized assessment of functional abilities for use with individuals (ages 0 – 90+) having intellectual and developmental disabilities. The Domain Level Parent/Caregiver Form is administered as a semi-structured interview with the parent or caregiver and focuses on home and family- life behavior. Adaptive functioning within three domains, Communication, Daily Living Skills, and Socialization, are assessed.^{1 2}
- d) The Reiss Screen for Maladaptive Behavior(Reiss 1994) was created specifically for use with individuals with ID to screen for possible depression, psychosis and behavior problems. Caregivers are asked to rate 38 symptoms of psychiatric behavior as being “No Problem”, “Problem”, or “Major Problem” for the individual in question. Behaviors are categorized into 8 sub-domains reflecting (a) aggressive behavior, autistic-like behavior, (c) psychosis, (d) paranoia, (e)/(f) behavioral and physical symptoms of depression, (g) dependent personality disorder, and (h) avoidant disorder.
- e) The Neuropsychiatric Inventory (NPI)(Cummings et al. 1994) is a validated instrument used to assess neuropsychiatric symptoms in dementia. The questionnaire asks caregivers to retrospectively rate symptoms in 12 neuropsychiatric domains: delusions, hallucinations, agitation/ aggression, dysphoria/depression, anxiety, euphoria/elation, apathy/indifference, disinhibition, irritability/lability, aberrant motor behaviors, nighttime behavioral disturbances, and appetite/ eating disturbance.²

Health Status and Life Events

- a) Demographics Questionnaire; includes year of participant’s birth; sex; ethnicity and race; primary language; age of mother and of father at participant’s birth; handedness; primary residence; level of independence; activities (e.g., vocational training program, working in the community, postsecondary education program).
- b) Medical Health History Questionnaire; includes Psychiatric conditions (e.g., depressive disorder, anxiety disorder); Neurological conditions (e.g., dementia, stroke, seizure disorder); Eyes/ears/nose/throat (e.g., cataracts, visual impairment, hearing impairment); Cardiovascular conditions (e.g., congenital heart disease, pacemaker); Respiratory conditions (e.g., asthma, pneumonia, sleep apnea); Allergies; Dermatologic/Connective tissue conditions (e.g., alopecia, rosacea); Musculoskeletal conditions (e.g., osteoporosis; gout); Endocrine/metabolic conditions (e.g., diabetes, Vitamin B12 deficiency); Menstrual information (e.g., menopausal); Hormone replacement information; Gastrointestinal conditions

(e.g., celiac disease, irritable bowel syndrome); Hematopoietic conditions (e.g., anemia); Cancer; Surgery; Covid-19; and Other conditions;

- c) **Stress Index:** Caregivers were asked about the presence or absence of significant life events and were asked to rate how the participant handled the experience.²

1.5 Special Cautions When Using this Dataset

Modified Cued Recall Test. Two sets of stimuli images (Set A/ Version 1 or Set B/ Version 2) were used. The set used varied across site and data collection cycle. Analyses conducted thus far suggest that set/version administered influences participant performance. The following variables are unique to the ABC-DS U19 study: `crset`, `ift1r`, `ift2r`, `ift3r`, `tifrrsps`, `agfrin1`, `agfrin2`, `agfrin3`, `tafi`, `c1t1`, `c1t2`, `c1t3`, `c2t1`, `c2t2`, `c2t3`, `c3t1`, `c3t2`, `c3t3`, `crc1`, `crc2` and `crc3`.

Down Syndrome Mental Status Examination (DSMSE): A component of the DSMSE, the Shoebox Memory Test, was unique to the ADDS U01 and ABC-DS U19 studies; thus, sample sizes for the `colshbx` and `dsto2` will be smaller than those for other DSMSE variables. Similarly, variables `colshbxtime`, `dsmeltime`, `colmemscr` and `nonmemory` are unique to the ABC-DS U19 study; thus, sample sizes for these variables will be smaller than those for other variables.

Modified Cats and Dogs Test: The variables, `nttrain` and `strain`, are new and unique to the ABC-DS U19 study.

NTG-Early Detection Screen for Dementia (NTG-EDSD): The Caregiver Concern Questions, variables `ntspcon` and `ntspconage`, are unique to the ADDS U01 and ABC-DS U19 studies.

Purdue Pegboard Test: Variables `pp_dh_tr` and `pp_ndh_tr` are unique to the ABC-DS U19 study. The variable `de_handedness` can be used to determine participants' dominant hand (1 = Right-handed, 2 = Left-handed, 3 = Ambidextrous).

Stress Index: The variable `si_victim` is a new variable unique to the ABC-DS U19 study.

Tinetti Gait Assessment: The following variables are unique to the ABC-DS U19 study: `tbagait`, `tbaslh`, `tbafe`, `tbass`, `tbasc`, `tbap`, `tbat`, and `tbawt`.

Verbal Fluency- McCarthy Category Fluency Test. The "Exceptional Scoring Method" is used. Variables `vffdexpr` and `vfanexpr` are unique to the ADDS U01 and ABC-DS U19 studies; the variable `vftotexpr` is unique to the ABC-DS U19 study.

1.6 Protocol for Sibling Controls

Approximately 50 sibling controls were recruited and underwent the same procedures with the exception of no caregiver assessments and a small cognitive screening battery. In addition, during the 2015-2020 U01 grant, only baseline blood draws were conducted. Sibling controls provided medical histories and were screened using the AD8(Galvin et al. 2005) (a brief informant interview to detect dementia) and the Montreal Cognitive Assessment (MoCA).(Nasreddine et al. 2005) The two screens were used to simply ensure that the siblings did not have MCI/Dementia and are not included in the ABC-DS database.

2 Summary

In summary, this book has no content whatsoever.

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